Pharmacies at the Hospital Gate: Drug Markup Regulation and Spatial Concentration

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Background and Data

Institutional Background

China's pharmaceutical reform had two key components. These components have potentially ambiguous implications for retail pharmacies.

- ↑ **Separated prescribing from dispensing:** allowing patients to fill prescriptions at retail pharmacies rather than purchasing drugs exclusively from hospitals.

- ↓ **Reduced drug prices in public hospitals:** by requiring them to sell medications at procurement cost, eliminating the previous markup on drug sales.

Reform Rollout and Scope

The reform began with county-level public-hospital pilots and expanded nationwide in stages.

- In 2012, the first pilot covered roughly 300 counties or county-level cities.
- By September 2017, public hospitals nationwide had removed drug mark-ups, except for designated traditional Chinese medicine products.

Data

- Business-Registration: Hospitals records, ownership, opening and cancellation dates, and geographic coordinates.
- DZDP: Pharmacies records and locations.
- Official government documents and related reports: county policy years.
- 2,512 counties and 110,547 hospital-year observations covering 2013–2018.
- Counties appear in every year, before the centralized drug procurement policy announced in November 2018 and piloted from January 2019.

Framework and Empirical Strategy

Conceptual Framework

A pharmacy at distance d from hospital h serves residential demand $A(d)$ and hospital-related demand $F_h s(d)\theta$. With margin m and fixed costs and rent $c(d)$,

$$\pi_h(d) = m\big[A(d) + F_h s(d)\theta\big] - c(d). \tag{1}$$

The reform raises the outside prescription share from θ to $\theta + \delta$ and lowers the retail margin from m to $m - \kappa$. Holding competition and rents fixed,

$$\Delta\pi_h(d) = F_h s(d)\big[(m - \kappa)\delta - \kappa\theta\big] - \kappa A(d). \tag{2}$$

- The prescription gain scales with patient volume F_h and declines with distance through $s(d)$.
- The residential margin loss does not depend on the hospital.
- Pharmacy entry should shift toward the gates of large hospitals and away from surrounding neighborhoods.

Hospital-Level Sample, Treatment, and Distance Rings

- 28,989 hospitals; 110,547 hospital-years, 2013–2018, in the balanced set of 2,512 counties.
- All public hospitals, including public TCM hospitals, are treated from the first calendar year after county adoption. Private hospitals are comparisons.
- Three disjoint bands: gate (0–200 m), near ring (200–500 m), outer ring (500–1,000 m). Also report the 0–1,000 m total.

$$NetEntry_{htr} = Stock_{htr} - Stock_{h,t-1,r}. \tag{3}$$

Net entry is undefined in the first hospital-year and after a gap. The preferred lag-valid regression sample has 74,924 observations; stock regressions have 103,766.

Summary Statistics

	N	Mean	SD	Min	Max
Public hospital	74,924	0.222	0.416	0	1
Public TCM hospital	74,924	0.043	0.203	0	1
Grade 1 hospital	68,254	0.696	0.460	0	1
Grade 2 hospital	68,254	0.223	0.416	0	1
Grade 3 hospital	68,254	0.081	0.272	0	1
Pharmacies within 200 m	74,924	1.288	1.971	0	56
Pharmacies within 500 m	74,924	5.190	6.038	0	86
Pharmacies within 800 m	74,924	10.961	11.799	0	121
Pharmacies within 1,000 m	74,924	15.464	16.236	0	161
Net entry within 200 m	74,924	0.133	0.885	-44	16
Net entry within 500 m	74,924	0.512	2.370	-67	25
Net entry within 800 m	74,924	1.072	4.483	-86	42
Net entry within 1,000 m	74,924	1.502	6.176	-113	56
Distance to nearest pharmacy (km)	63,041	0.226	0.216	0.00	1.13

Notes: The sample contains the hospital-years in the net-entry regressions. Pharmacy counts are active Dianping pharmacies within each radius, and net entry is the annual change in that count. Tier shares use hospitals with a classified tier.

Baseline Model

$$Y_{hctr} = \beta_r\big(Public_h \times Post_{ct}\big) + \alpha_h + \lambda_{ct} + \varepsilon_{hctr}. \tag{4}$$

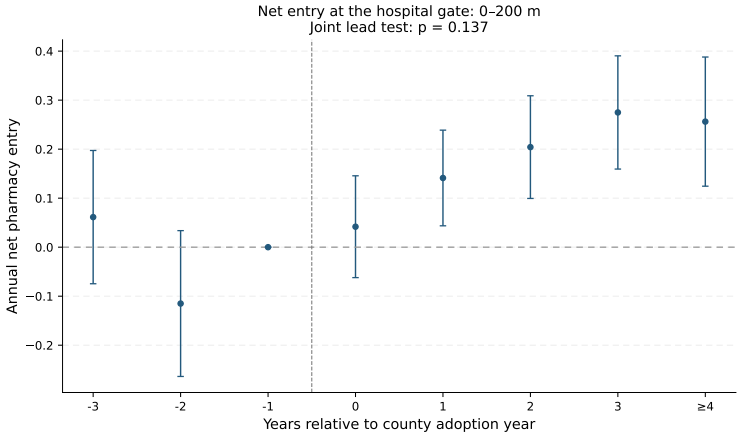
- Y_{hctr} is net pharmacy entry or the pharmacy stock in distance band r around hospital h in county c and year t.
- $Post_{ct} = \mathbb{1}\{t > PolicyYear_c\}$.
- Hospital FE α_h absorb fixed hospital and neighborhood characteristics.
- County-by-year FE λ_{ct} compare public and private hospitals in the same county-year and absorb county-wide reform and local shocks.
- County-clustered standard errors. Parallel trends are required for the public–private difference in net entry.

Event-Study Specification and Timing

$$Y_{hctr} = \sum_{k \neq -1} \beta_{kr} Public_h \mathbb{1}\{t - T_c = k\} + \alpha_h + \lambda_{ct} + \varepsilon_{hctr}. \tag{5}$$

- T_c is the last pre-reform year, so $k = 0$ is the first reformed year and $k = -1$ is the omitted reference.
- Endpoints bin four or more years before and after the reform.
- Joint lead tests use the estimable pre-reform indicators.
- Confidence intervals and joint tests use standard errors clustered by county.

Net Pharmacy Entry at the Gate: Event Study



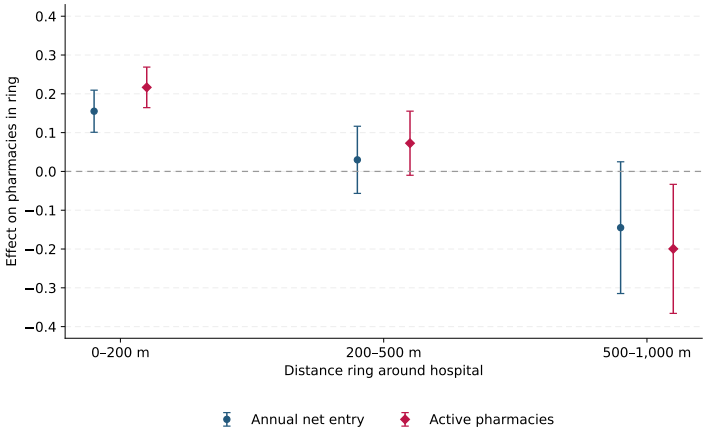
The vertical line separates the omitted pre-period from the adoption year. The joint test of the pre-reform coefficients has $p = 0.137$. The test does not establish parallel trends.

Effects of the Zero-Markup Reform on Pharmacies by Distance

	Gate 0–200 m (1)	Near ring 200–500 m (2)	Outer ring 500–1,000 m (3)	Total 0–1,000 m (4)
Panel A. Annual net entry				
Public × Post	0.155***	0.030	-0.145*	0.040
	(0.028)	(0.044)	(0.087)	(0.114)
Mean of outcome	0.133	0.379	0.989	1.502
Pre-trend p-value	0.137	0.795	0.936	0.785
Observations	74,924	74,924	74,924	74,924
Panel B. Active pharmacies				
Public × Post	0.217***	0.073*	-0.200**	0.090
	(0.027)	(0.042)	(0.085)	(0.112)
Mean of outcome	1.273	3.782	9.892	14.947
Observations	103,766	103,766	103,766	103,766

Notes: All regressions include hospital and county-by-year fixed effects. Standard errors clustered by county are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Effects on Pharmacies by Distance Ring



Circles: annual net entry. Diamonds: active pharmacies. Bars show 95% confidence intervals; effects are estimated directly on each ring.

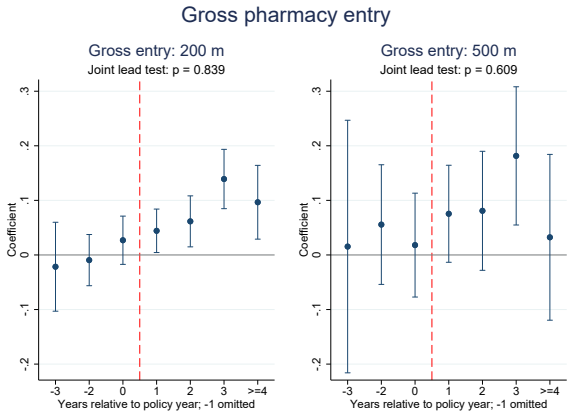
Gross Pharmacy Entry and Exit

	200 m	500 m	800 m	1,000 m
Gross entry	0.043***	0.068**	0.006	-0.052
	(0.014)	(0.030)	(0.048)	(0.058)
Gross exit	-0.027**	-0.043*	-0.001	-0.018
	(0.011)	(0.026)	(0.046)	(0.060)
Outcome mean: entry	0.227	1.019	2.215	3.163
Outcome mean: exit	0.122	0.567	1.243	1.785
Observations	93,265	93,265	93,265	93,265

The gate effect combines more pharmacy appearances with fewer disappearances. Neither margin changes at wider radii.

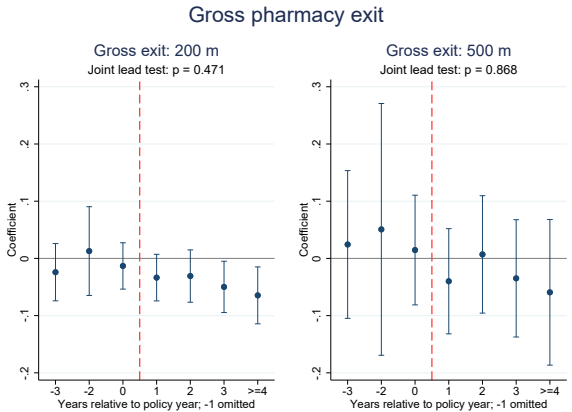
Notes: Entry is a `shop_id` observed in year t but not $t-1$; exit is defined conversely. These are platform appearance and disappearance measures. Models absorb hospital and county-by-year fixed effects; standard errors are clustered by county. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Gross Pharmacy Entry: Event Studies



The omitted period is one year before the policy year. Each panel reports the joint test of the estimable pre-reform coefficients; bars show 95% confidence intervals.

Gross Pharmacy Exit: Event Studies



The omitted period is one year before the policy year. Each panel reports the joint test of the estimable pre-reform coefficients; bars show 95% confidence intervals.

Stacked DID: Effects by Distance Ring

	Gate 0–200 m	Near ring 200–500 m	Outer ring 500–1,000 m	Total 0–1,000 m
Annual net entry	0.157***	0.031	-0.137	0.051
	(0.028)	(0.045)	(0.090)	(0.117)
Pre-trend p-value	0.102	0.948	0.904	0.910
Active pharmacy stock	0.222***	0.075*	-0.191**	0.107
	(0.027)	(0.043)	(0.087)	(0.114)

The mutually exclusive rings reproduce the baseline spatial pattern. Net entry rises at the gate and falls in the outer ring, leaving the 1 km total small and imprecise. The stock estimates show the same reallocation.

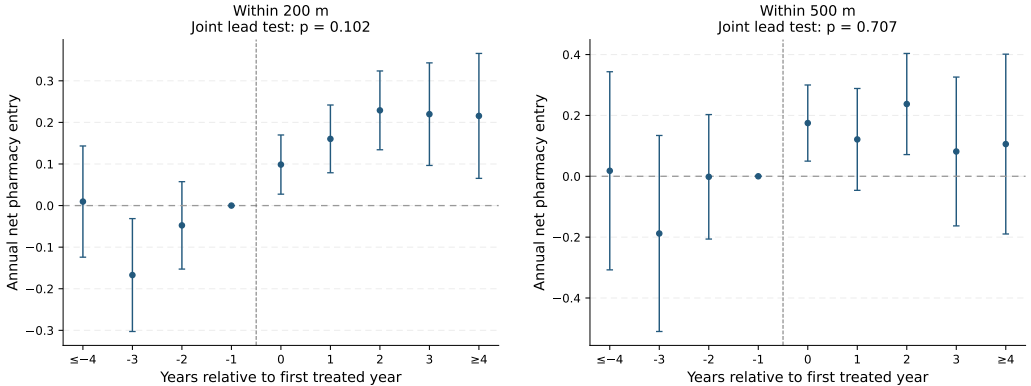
Notes: Each stack pairs one public-hospital cohort with private hospitals in counties containing both groups. Models absorb stack-by-hospital and stack-by-county-by-year fixed effects. County-clustered standard errors are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Robustness: Stacked DID with County-by-Year Fixed Effects

	200 m	500 m	800 m	1,000 m
Annual net entry	0.157***	0.189***	0.103	0.051
	(0.028)	(0.054)	(0.087)	(0.117)
Active pharmacy stock	0.222***	0.297***	0.172**	0.107
	(0.027)	(0.054)	(0.086)	(0.114)
Net-entry lead-test p-value	0.102	0.707	0.689	0.910

Each stack pairs one public-hospital treatment cohort with non-public hospitals in counties containing both groups. The model absorbs stack-by-hospital and stack-by-county-by-year fixed effects; standard errors are clustered by county. Net entry rises within 200 m and 500 m, then becomes small and imprecise at wider radii. Stock estimates are supporting evidence because their lead tests reject.

Stacked DID: Net-Entry Event Studies



The omitted period is one year before treatment. The vertical line marks the transition into treatment. Joint lead-test p-values are 0.102 within 200 m and 0.707 within 500 m; bars show 95% confidence intervals.

Robustness: Estimators and Treatment Definitions

	200 m (1)	500 m (2)	800 m (3)	1,000 m (4)
Panel A. Alternative estimators and treatment definitions				
(1) Baseline	0.155***	0.185***	0.097	0.040
	(0.028)	(0.053)	(0.085)	(0.114)
Pre-trend p-value	0.137	0.519	0.455	0.785
(2) Cohort-saturated TWFE	0.147***	0.182***	0.098	0.018
	(0.028)	(0.055)	(0.085)	(0.114)
(3) Public non-TCM treated	0.166***	0.197***	0.063	-0.035
	(0.029)	(0.054)	(0.087)	(0.112)
Panel B. Public non-TCM and public TCM hospitals				
Public non-TCM × Post	0.172***	0.204***	0.078	-0.011
	(0.030)	(0.055)	(0.089)	(0.116)
Public TCM × Post	0.072	0.089	0.191	0.299
	(0.055)	(0.105)	(0.163)	(0.218)

Notes: Annual net pharmacy entry within each radius. All models include hospital and county-by-year fixed effects. County-clustered standard errors are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Placebo: Non-Pharmacy Wellness Establishments

	0–200 m	200–500 m	500–1,000 m	0–1,000 m
Net entry	-0.026	-0.038	-0.005	-0.069
	(0.017)	(0.033)	(0.056)	(0.073)
Stock	0.020	0.039	-0.158**	-0.099
	(0.013)	(0.030)	(0.062)	(0.083)

- Net entry does not increase in any distance band; the total stock within 1 km also shows no significant increase.
- The negative stock estimate at 500–1,000 m also provides no evidence of expansion. Together, the results do not support a broad rise in health demand or ordinary commercial growth around hospitals.

Notes: Outcomes count non-pharmacy wellness establishments. Net entry is the annual change in the count; stock is the annual level. All models absorb hospital and county-by-year fixed effects. Standard errors clustered by county are in parentheses. ** $p < 0.05$.

Concentration around Tertiary Hospitals

Hospital-Tier Model and Formal Difference Tests

$$Y_{hct}^{(r)} = \beta_{1r} D_{hct} + \delta_{2r} D_{hct} \mathbb{1}\{G_h = 2\} + \delta_{3r} D_{hct} \mathbb{1}\{G_h = 3\} + \alpha_h + \lambda_{c \times t} + \varepsilon_{hct}^{(r)}.$$

- $D_{hct} = Public_h \times Post_{ct}$; G_h is the final hospital-tier classification.
- Total effects: Grade 1 $= \beta_{1r}$; Grade 2 $= \beta_{1r} + \delta_{2r}$; Grade 3 $= \beta_{1r} + \delta_{3r}$.
- The formal Grade 3–Grade 1 contrast is δ_{3r}, computed from the pooled model with its full covariance matrix.
- Tier results use cumulative radii, matching Table 4 of the paper.

Hospital Tier: Annual Net Pharmacy Entry

	200 m	500 m	800 m	1,000 m
Grade 1	0.031	-0.037	-0.214*	-0.279*
	(0.034)	(0.076)	(0.119)	(0.160)
Grade 2	0.116***	0.123	0.048	-0.006
	(0.042)	(0.080)	(0.124)	(0.155)
Grade 3	0.323***	0.507***	0.494***	0.464**
	(0.053)	(0.097)	(0.153)	(0.192)
Grade 3 − Grade 1	0.292***	0.544***	0.708***	0.743***
	(0.059)	(0.116)	(0.177)	(0.220)
Observations	67,948	67,948	67,948	67,948

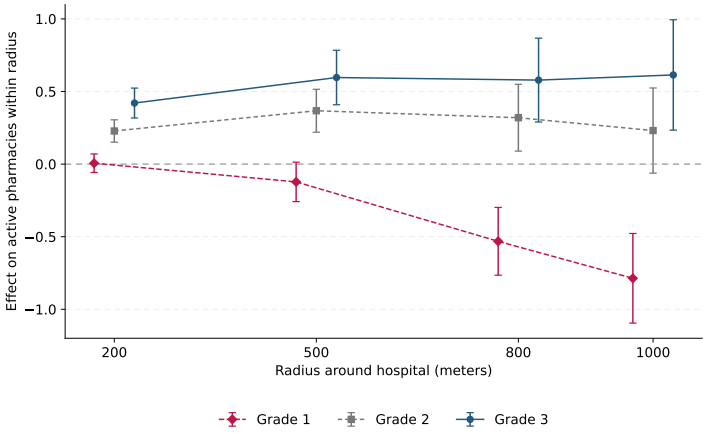
Classified-tier sample; all public hospitals treated. Hospital and county-by-year FE; county-clustered SE. Tier includes administrative grades and classifications inferred from hospital scale.

Hospital Tier: Active Pharmacy Stocks

	200 m	500 m	800 m	1,000 m
Grade 1	0.006	-0.123*	-0.532***	-0.786***
	(0.032)	(0.069)	(0.119)	(0.157)
Grade 2	0.228***	0.367***	0.319***	0.231
	(0.039)	(0.075)	(0.117)	(0.150)
Grade 3	0.420***	0.596***	0.579***	0.614***
	(0.053)	(0.096)	(0.147)	(0.194)
Grade 3 − Grade 1	0.414***	0.719***	1.110***	1.400***
	(0.058)	(0.107)	(0.173)	(0.229)
Observations	94,717	94,717	94,717	94,717

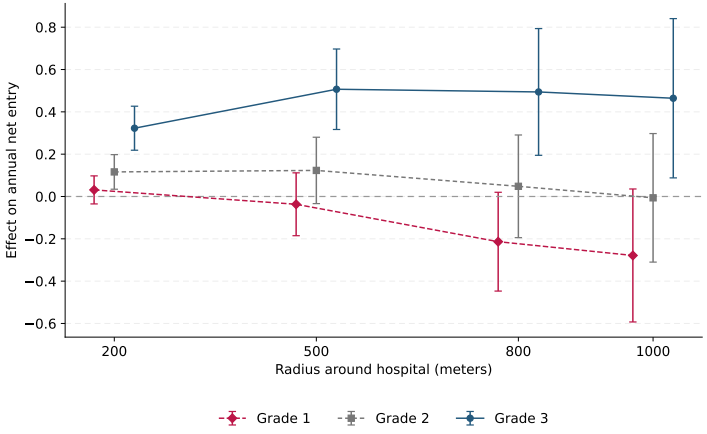
Each radius is estimated in one pooled interaction model. The last row formally tests the Grade 3–Grade 1 difference. County-clustered SE in parentheses.

Effects on Pharmacy Stocks by Hospital Tier and Radius



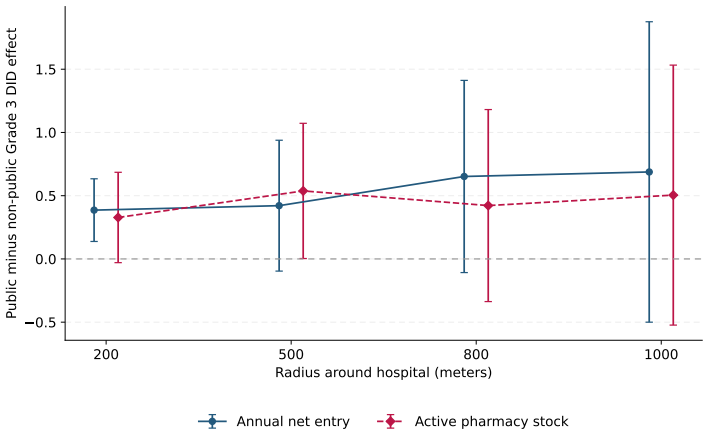
Cumulative radii; total effects from the pooled model. Bars show 95% confidence intervals. Horizontal offsets improve readability.

Effects on Net Entry by Hospital Tier and Radius



Grade 3 shows positive net entry; Grade 1 estimates turn negative at wider radii. Bars show 95% confidence intervals.

Public versus Non-Public Grade 3 Hospitals



The coefficients are within-tier DID estimates: public Grade 3 hospitals are treated and non-public Grade 3 hospitals are the untreated comparison group. Bars show 95% confidence intervals.

Spatial Concentration and Access

- Within hospital neighborhoods: entry at the gate offsets losses in the outer ring, leaving the 1 km total imprecise.
- Across hospitals: supply gains concentrate around Grade 3 public hospitals, while wider-radius stocks fall around Grade 1 hospitals.
- Patients filling hospital prescriptions may gain convenient outside options; residents relying on neighborhood pharmacies may face longer trips.
- Location counts do not identify individual pharmacy relocations or measure patient welfare directly. These remain interpretation and future measurement tasks.

Residential Proximity and Population-Weighted Access

Pharmacy–community distance

- Residential communities come from annual Amap POI records for 2013–2018.
- For each pharmacy, we calculate the distance to its nearest residential community.
- Hospital-year means use cumulative radii (200, 500, and 1,000 m) and mutually exclusive rings (0–200, 200–500, and 500–1,000 m).

Spatial inequality in access

- WorldPop annual population rasters provide approximately 100 m residential demand cells.
- 2SFCA first divides each pharmacy by population in its 500 m catchment, then sums nearby ratios for each grid cell.
- Population-weighted Gini is calculated within 500 m, 2 km, and 3 km of each hospital. The 2 km radius is the main specification.
- Annual population weights are the main measure. Fixed 2013 weights hold population locations constant.

Sources: Amap POI (residential communities); WorldPop annual gridded population, 2013–2018. Hospital distance is measured from the AOI boundary when available and from the hospital point otherwise.

Pharmacies Moved Closer to Communities in the 200–500 m Ring

Hospital area	DID effect (m)	Standard error	p-value	Observations
Stock: within 200 m	-49.51	41.96	0.238	29,508
Stock: within 500 m	-55.08*	30.68	0.073	50,036
Stock: within 1,000 m	-44.51	30.04	0.139	59,366
Ring: 0–200 m	-49.51	41.96	0.238	29,508
Ring: 200–500 m	-58.51**	26.20	0.026	46,205
Ring: 500–1,000 m	-18.33	18.36	0.318	54,688

- Negative coefficients indicate that pharmacies around treated public hospitals became closer to their nearest residential community relative to pharmacies around private hospitals.
- The result is concentrated in the 200–500 m ring: the mean pharmacy–community distance fell by about 59 m.

Notes: Outcomes are annual changes in the hospital-year mean distance. Models absorb hospital and county-by-year fixed effects; standard errors are clustered by county. $^{*}p < 0.10$, $^{**}p < 0.05$.

Relocating Pharmacies: Distances to Communities and Hospitals

Hospital area	Δ nearest community (m)	Δ nearest hospital (m)	Observations
Stock: within 200 m	-28.42	-98.00*	360
	(29.91)	(55.72)	
Stock: within 500 m	-35.86	-63.29	5,195
	(28.71)	(47.48)	
Stock: within 1,000 m	-5.19	30.71	19,541
	(16.88)	(31.19)	
Ring: 0–200 m	-28.42	-98.00*	360
	(29.91)	(55.72)	
Ring: 200–500 m	-5.50	-29.48	3,260
	(18.52)	(36.57)	
Ring: 500–1,000 m	-20.38	44.12	13,063
	(23.16)	(40.34)	

- At the hospital gate (0–200 m), relocating pharmacies moved about 98 m closer to their nearest hospital relative to the control group.
- Community-distance effects are negative and insignificant, so the results do not support relocation away from residential communities.

Notes: A relocating pharmacy is the same `shop_id` moving at least 100 m between adjacent years. Outcomes are destination-minus-origin distance changes, evaluated against the same destination-year POI sets. DID models absorb hospital and county-by-year fixed effects; standard errors (in parentheses) are clustered by county. $^{*}p < 0.10$.

Population-Relative Pharmacy-Share Gini: No Robust Reform Effect

	500 m	2 km (main)	3 km
Annual population weights	-0.00035	0.00002	-0.00000
	(0.00035)	(0.00008)	(0.00006)
Fixed 2013 population weights	-0.00036	0.00002	0.00001
	(0.00035)	(0.00008)	(0.00006)
Observations	72,956	90,784	93,177
Lead-test p-value (annual weights)	0.534	0.656	0.117

- The Gini compares each grid cell's pharmacy share with its population share; zero indicates proportional spatial matching and larger values indicate greater mismatch.
- The 2 km main estimate is economically tiny and statistically insignificant. Results are similar when population is fixed at its 2013 distribution.

Notes: WorldPop native population cells are retained, including populated cells with zero pharmacies. Cells are ranked by pharmacies per resident. Models absorb hospital and county-by-year fixed effects; standard errors (in parentheses) are clustered by county. No coefficient is statistically significant at the 10% level.

2SFCA Results: No Increase in Spatial Inequality

	500 m	2 km (main)	3 km
Annual population weights	-0.0039*	-0.0017*	-0.0011
	(0.0023)	(0.0009)	(0.0007)
Fixed 2013 population weights	-0.0038*	-0.0015	-0.0007
	(0.0023)	(0.0009)	(0.0007)
Outcome mean (annual weights)	0.3420	0.6016	0.6576
Observations	84,046	92,136	93,920
Lead-test p-value	0.624	0.823	0.373

- All estimates are negative. The data provide no evidence that the reform increased inequality in population-adjusted pharmacy accessibility.
- In the 2 km main specification, Gini falls by 0.0017, about 0.3% of its mean. The lead test provides no evidence of differential pre-trends.

Notes: 2SFCA accessibility is the sum of pharmacy-to-population ratios within a 500 m floating catchment, scaled per 10,000 residents. The outcome is its population-weighted Gini within each hospital radius. Models absorb hospital and county-by-year fixed effects; standard errors are clustered by county. $^{*}p < 0.10$.

Heterogeneity

County Characteristics and the Net-Entry Effect

DID × standardized characteristic	200 m	500 m	800 m	1,000 m
Population age 65+ share	0.027	0.092	0.119	0.276**
	(0.027)	(0.059)	(0.094)	(0.115)
Urban population share	0.045	0.094*	0.097	0.162
	(0.028)	(0.055)	(0.091)	(0.118)
Average years of education	0.065**	0.128**	0.160	0.226*
	(0.029)	(0.065)	(0.104)	(0.131)
Log population density	0.062**	0.138**	0.179*	0.234**
	(0.027)	(0.057)	(0.092)	(0.118)
Observations	67,546	67,546	67,546	67,546

The response is stronger in denser and more educated counties. Urbanization is weaker, while the elderly-share interaction appears only at 1,000 m.

Notes: Each row comes from a separate model. Characteristics are fixed at their 2010 census values and standardized in the estimation sample. Models include the DID main effect, hospital FE, and county-by-year FE; standard errors are clustered by county. The correlated characteristics and multiple tests make these exploratory results. $^{*}p < 0.10$, $^{**}p < 0.05$.

General and Specialist Hospitals

Net-entry effect	200 m	500 m	800 m	1,000 m
General hospitals	0.227***	0.311***	0.206*	0.162
	(0.042)	(0.080)	(0.120)	(0.158)
Specialist hospitals	0.091*	0.010	-0.155	-0.351
	(0.051)	(0.104)	(0.178)	(0.229)
Specialist minus general	-0.136**	-0.301**	-0.360*	-0.513**
	(0.066)	(0.127)	(0.199)	(0.254)
Observations	55,570	55,570	55,570	55,570

Pharmacy growth is concentrated around general hospitals. Specialist-hospital effects are small or negative, and the formal specialist-minus-general contrasts are negative at every radius.

Notes: The pooled interaction model restricts the sample to hospitals classified as general or specialist. It includes hospital and county-by-year fixed effects; standard errors are clustered by county. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Conclusion

Main Findings

1. Hospital net entry rises in the 0–200m gate band and falls in the 500–1,000m outer ring; the 1km total is small and imprecise. Gains concentrate around Grade 3 hospitals.
2. At the gate, the reform increases gross pharmacy entry and reduces gross exit. Neither margin changes at wider radii.
3. The pharmacy response is stronger in denser and more educated counties, but correlated characteristics and multiple tests make the interaction results exploratory.
4. Net entry increases around general hospitals and shows no clear positive response around specialist hospitals.
5. Population-adjusted access measures provide no evidence that spatial inequality increased after the reform.